

MTH2021 – Complex Analysis

Part 1

January 20, 2023

Tchavdar Todorov
email: t.todorov@qub.ac.uk

Based on materials generously provided by Patrick Norrington

Contents

1	Complex numbers	3
2	Complex plane	4
3	Modulus and argument	5
4	Point at infinity	7
5	Equality, negative	8
6	Addition/subtraction, multiplication	9
7	Division	9
8	Complex conjugate	11
9	Euler's formula and polar form	13
10	De Moivre's theorem	15
11	Integer powers	18
12	n -th roots	19
13	Inequalities	20
14	Polynomial equations	22
15	Complex functions	23
16	Neighbourhood and annulus	24
17	Limit	25
18	Continuity	25

19 Derivative	27
20 Analytic functions	29
21 Cauchy-Riemann equations	30
22 Exponential function	32
23 Circular functions	33
24 Hyperbolic functions	34
25 Entire functions	35
26 Logarithm	36
27 Complex powers	37
28 Branch points and cuts	38
29 Harmonic functions	40
30 Series	42
31 Convergence tests	43

1 Complex numbers

A complex number z is written as

$$z = x + iy$$

where x and y are real numbers, and i is the imaginary unit.

The imaginary unit is such that $i^2 = -1$.

Complex algebra is like real algebra with i^2 set to -1 when it occurs.

Definition

$$\begin{array}{ll} \text{Real part of } z: & \operatorname{Re} z = x \\ \text{Imaginary part of } z: & \operatorname{Im} z = y \end{array}$$

The real number $z = x$ is a complex number with $\operatorname{Im} z = 0$.

The imaginary number $z = iy$ is a complex number with $\operatorname{Re} z = 0$.

0, 1 and i are all complex numbers.

2 Complex plane

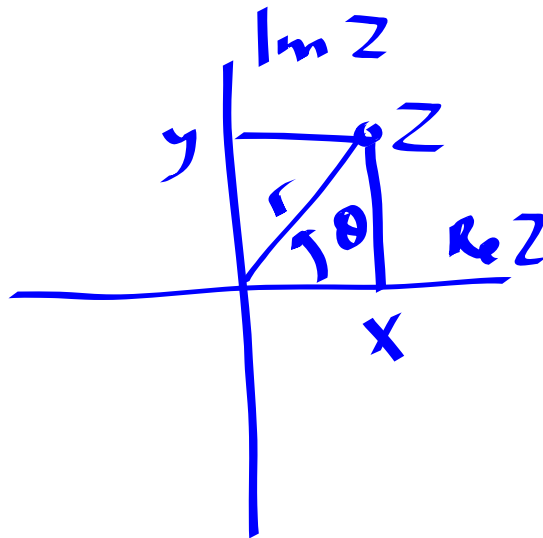
Complex numbers can be represented by points on a plane, known as the complex plane (or Argand diagram). The complex number $z = x + iy$ corresponds to a point with Cartesian coordinates (x, y) .

For points on a plane we also use polar coordinates (r, θ) where

$$x = r \cos \theta \quad \text{and} \quad y = r \sin \theta$$

so that

$$r = \sqrt{x^2 + y^2} \geq 0 \quad \text{and} \quad \tan \theta = \frac{y}{x} \quad \text{with} \quad x \neq 0$$



Definition

$$\begin{array}{ll} \text{Modulus of } z: & |z| = r \\ \text{Argument of } z: & \arg z = \theta \end{array}$$

Note that $\arg 0$ is undefined.

3 Modulus and argument

The modulus of z is the distance of the point from the origin.

If θ is the argument of z then so is

$$\theta + 2k\pi \quad \text{where} \quad k = 0, \pm 1, \pm 2, \dots$$

Therefore $\arg z$ is a multi-valued function of z .

It is convenient to define a single-valued function, called the principal value of $\arg z$, and denoted by $\text{Arg } z$. $\text{Arg } z$ is the unique value such that

$$-\pi < \text{Arg } z \leq \pi$$

Then

$$\arg z = \text{Arg } z + 2k\pi \quad \text{where} \quad k = 0, \pm 1, \pm 2, \dots$$

We work out $\text{Arg } z$ as follows.

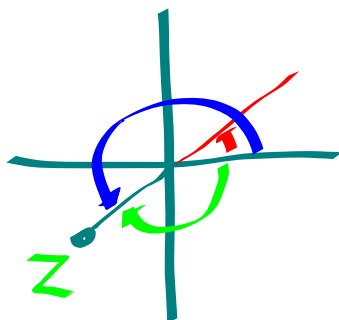
a. If $x > 0$, then $\text{Arg } z = \text{Arctan}(y/x)$.

Arctan (the principal value of the inverse tangent function) is a value in the range $(-\pi/2, \pi/2)$.

b. If $x = 0$, then $\text{Arg } z = \pm\pi/2$ where the sign is the sign of y .

c. If $x < 0$ and $y \neq 0$, then $\text{Arg } z = \text{Arctan}(y/x) \pm \pi$ where the sign is the sign of y .

d. If $x < 0$ and $y = 0$, then $\text{Arg } z = \pi$.



Examples

$z = 2 \Rightarrow x = 2, y = 0 \Rightarrow |z| = \sqrt{x^2 + y^2} = 2$. For the argument, we have case **a.**: $\text{Arg } z = \text{Arctan}(y/x) = \text{Arctan } 0 = 0$.

$z = 1 + i \Rightarrow x = 1, y = 1 \Rightarrow |z| = \sqrt{x^2 + y^2} = \sqrt{2}$. For the argument, we have case **a.**: $\text{Arg } z = \text{Arctan}(y/x) = \text{Arctan } 1 = \pi/4$.

$z = i \Rightarrow x = 0, y = 1 \Rightarrow |z| = \sqrt{x^2 + y^2} = 1$. For the argument, we have case **b.**: $\text{Arg } z = \pi/2$.

$z = -1 + i \Rightarrow x = -1, y = 1 \Rightarrow |z| = \sqrt{x^2 + y^2} = \sqrt{2}$. For the argument, we have case **c.**: $\text{Arg } z = \text{Arctan}(y/x) + \pi = \text{Arctan}(-1) + \pi = -\pi/4 + \pi = 3\pi/4$.

$z = -3 \Rightarrow x = -3, y = 0 \Rightarrow |z| = \sqrt{x^2 + y^2} = 3$. For the argument, we have case **d.**: $\text{Arg } z = \pi$.

$z = -\sqrt{3}/2 - i/2 \Rightarrow x = -\sqrt{3}/2, y = -1/2 \Rightarrow |z| = \sqrt{x^2 + y^2} = 1$. For the argument, we have case **c.**: $\text{Arg } z = \text{Arctan}(y/x) - \pi = \text{Arctan}(1/\sqrt{3}) - \pi = \pi/6 - \pi = -5\pi/6$.

$z = \sqrt{3}/2 - i/2 \Rightarrow x = \sqrt{3}/2, y = -1/2 \Rightarrow |z| = \sqrt{x^2 + y^2} = 1$. For the argument, we have case **a.**: $\text{Arg } z = \text{Arctan}(y/x) = \text{Arctan}(-1/\sqrt{3}) = -\pi/6$.

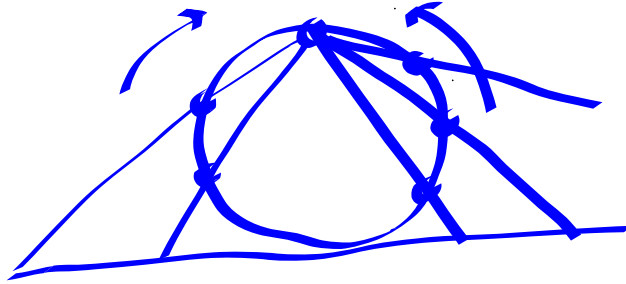
Show the various values of z above on the Argand diagram.

4 Point at infinity

There is a one-to-one correspondence between the complex plane and the surface of a sphere. To see this, place a sphere resting on the plane with the south pole at the origin. The polar axis is perpendicular to the plane. For a point P on the plane draw a line joining P and the north pole. This intersects the surface of the sphere at a point P' .

As P moves far from the origin P' approaches the north pole. The north pole corresponds to infinity. It is a single point called the point at infinity.

An application is in investigating the behaviour of functions $f(z)$ as $z \rightarrow \infty$. To handle this we transform variable to $w = 1/z$ and consider the behaviour of $f(1/w)$ near $w = 0$.



5 Equality, negative

Equality

$$\begin{aligned} z_1 = z_2 &\Rightarrow x_1 + iy_1 = x_2 + iy_2 \\ &\Leftrightarrow x_1 = x_2 \quad \text{and} \quad y_1 = y_2 \end{aligned}$$

Two complex numbers are equal if and only if their real parts are equal and separately their imaginary parts are equal.

This is analogous to equality for vectors.

Negative

$$\begin{aligned} -z &= -(x + iy) \\ &= (-x) + i(-y) \end{aligned}$$

6 Addition/subtraction, multiplication

Addition/subtraction

$$\begin{aligned}z_1 \pm z_2 &= (x_1 + iy_1) \pm (x_2 + iy_2) \\&= (x_1 \pm x_2) + i(y_1 \pm y_2)\end{aligned}$$

This is analogous to vector addition in the plane.

Multiplication

$$\begin{aligned}z_1 z_2 &= (x_1 + iy_1)(x_2 + iy_2) \\&= x_1 x_2 + i x_1 y_2 + i y_1 x_2 + i^2 y_1 y_2 \\&= x_1 x_2 + i x_1 y_2 + i y_1 x_2 - y_1 y_2 \\&= (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2)\end{aligned}$$

It follows that $z_1 z_2 = z_2 z_1$ and $z_1(z_2 z_3) = (z_1 z_2)z_3 \equiv z_1 z_2 z_3$.

Example

$$z_1 = 2 - i \quad \text{and} \quad z_2 = -1 + 3i$$

$$\begin{aligned}z_1 + z_2 &= 1 + 2i \\z_1 - z_2 &= 3 - 4i\end{aligned}$$

$$\begin{aligned}z_1 z_2 &= (2 - i)(-1 + 3i) \\&= -2 + 6i + i - 3i^2 \\&= -2 + 6i + i + 3 = 1 + 7i\end{aligned}$$

7 Division

First we define the multiplicative inverse, z^{-1} , of a complex number $z \neq 0$, to be such that $zz^{-1} = z^{-1}z = 1$. It follows that z^{-1} , if it exists, is unique. We can now find it as follows. We have

$$\begin{aligned}z &= x + iy \\z(x - iy) &= (x + iy)(x - iy) \\&= x^2 + y^2 = (x^2 + y^2) + 0i\end{aligned}$$

We do not yet know how to divide by complex numbers, but we do know that real numbers have reciprocals. Let us, therefore, multiply both sides above by the complex number $\left(\frac{1}{x^2+y^2} + 0i\right)$:

$$\begin{aligned} z(x - iy) \left[\frac{1}{x^2 + y^2} + 0i \right] &= [(x^2 + y^2) + 0i] \left[\frac{1}{x^2 + y^2} + 0i \right] \\ z \left(\frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2} \right) &= 1 \end{aligned}$$

Hence

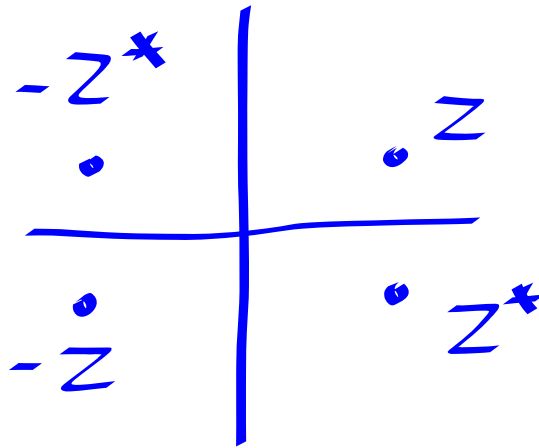
$$\begin{aligned} z^{-1} &= \frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2} \\ &= \frac{x - iy}{x^2 + y^2} \quad \text{for short} \end{aligned}$$

We now define division as multiplication by the multiplicative inverse:

$$\begin{aligned} \frac{z_1}{z_2} &\equiv z_1 z_2^{-1} = z_2^{-1} z_1 \\ &= (x_1 + iy_1) \left(\frac{x_2}{x_2^2 + y_2^2} - i \frac{y_2}{x_2^2 + y_2^2} \right) \\ &= \left(\frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} \right) + i \left(\frac{-x_1 y_2 + y_1 x_2}{x_2^2 + y_2^2} \right) \\ &= \frac{(x_1 + iy_1)(x_2 - iy_2)}{x_2^2 + y_2^2} \end{aligned}$$

8 Complex conjugate

The complex conjugate of a complex number $z = x + iy$ is $z^* = x - iy$. It corresponds to a reflection of the point across the real axis.



We have the relations:

$$\operatorname{Re} z = \frac{1}{2}(z + z^*)$$

$$\operatorname{Im} z = \frac{1}{2i}(z - z^*)$$

$$(z^*)^* = z$$

$$(z_1 \pm z_2)^* = z_1^* \pm z_2^*$$

$$(z_1 z_2)^* = z_1^* z_2^*$$

$$\left(\frac{z_1}{z_2}\right)^* = \frac{z_1^*}{z_2^*}$$

$$z z^* = x^2 + y^2 = |z|^2$$

$$z^{-1} = \frac{z^*}{z z^*}$$

Examples

$$z_1 = 1 + 2i \quad \text{and} \quad z_2 = 2 - i$$

$$z_1^* = 1 - 2i$$

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{(x_1 + iy_1)(x_2 - iy_2)}{x_2^2 + y_2^2} \\ &= \frac{(1 + 2i)(2 + i)}{4 + 1} = \frac{5i}{5} = i \end{aligned}$$

$$\begin{aligned} \frac{z_1}{z_2^*} &= \frac{(x_1 + iy_1)(x_2 + iy_2)}{x_2^2 + y_2^2} \\ &= \frac{(1 + 2i)(2 - i)}{4 + 1} = \frac{4 + 3i}{5} = \frac{4}{5} + \frac{3}{5}i \end{aligned}$$

9 Euler's formula and polar form

A complex number can be written as

$$\begin{aligned} z &= x + iy \\ &= r \cos \theta + ir \sin \theta \\ &= r (\cos \theta + i \sin \theta) \end{aligned}$$

Consider the combination $f(\theta) = \cos \theta + i \sin \theta$. It has the properties:

$$f(0) = 1$$

$$\begin{aligned} f(\theta)f(\phi) &= (\cos \theta + i \sin \theta)(\cos \phi + i \sin \phi) \\ &= (\cos \theta \cos \phi - \sin \theta \sin \phi) + i (\sin \theta \cos \phi + \cos \theta \sin \phi) \\ &= \cos (\theta + \phi) + i \sin (\theta + \phi) = f(\theta + \phi) \end{aligned}$$

$$\begin{aligned} \frac{df}{d\theta} &= -\sin \theta + i \cos \theta \\ &= i (\cos \theta + i \sin \theta) = i f(\theta) \end{aligned}$$

This is what we would expect the function $f(\theta) = \exp(i\theta)$ to do. We accordingly introduce the

$$\text{Euler formula :} \quad \cos \theta + i \sin \theta = \exp(i\theta)$$

on the understanding that for now the RHS is short-hand for the combination on the left, with the properties above.

Later - when we independently introduce the exponential function for complex variables - we will see that the Euler formula is indeed just a special case of $\exp z$.

Using the Euler formula we can write a complex number as

$$\begin{aligned} z &= r (\cos \theta + i \sin \theta) \\ &= r \exp(i\theta) \\ &= |z| \exp(i \arg z) \end{aligned}$$

This is the polar form of a complex number.

All values of $\arg z = \text{Arg } z + 2k\pi$, when used in the polar form, return the same z :

$$\begin{aligned} \exp(i \arg z) &= \exp[i(\text{Arg } z + 2k\pi)] \\ &= \exp(i \text{Arg } z) \exp(i2k\pi) \\ &= \exp(i \text{Arg } z) (\cos 2k\pi + i \sin 2k\pi) \\ &= \exp(i \text{Arg } z) \end{aligned}$$

We will soon see however that the $2k\pi$ can make a difference once we start to form further functions of z .

Example

$$\begin{aligned} z = 1 - i &= |z| \frac{z}{|z|} \\ &= \sqrt{2} \frac{1 - i}{\sqrt{2}} \\ &= \sqrt{2} [\cos(-\pi/4) + i \sin(-\pi/4)] = \sqrt{2} \exp(-i\pi/4) \end{aligned}$$

“by inspection”. Or if we prefer to follow the “recipe”:

$$z = 1 - i = x + iy$$

$$|z| = \sqrt{x^2 + y^2} = \sqrt{2}, \quad \text{Arg } z = \text{Arctan}(y/x) = \text{Arctan}(-1) = -\pi/4$$

$$z = |z| \exp(i \text{Arg } z) = \sqrt{2} \exp(-i\pi/4)$$

When $\theta = \pi$ the Euler formula gives the famous equation

$$\exp(i\pi) + 1 = 0$$

If we put $\exp(i\pi) = e^{i\pi}$, then this equation relates the numbers $e = 2.71828\dots$, $\pi = 3.14159\dots$, i , 1 and 0 .

10 De Moivre's theorem

We have seen that

$$(\cos \theta + i \sin \theta)(\cos \phi + i \sin \phi) = \cos (\theta + \phi) + i \sin (\theta + \phi)$$

Put $\phi = \theta$:

$$(\cos \theta + i \sin \theta)^2 = \cos (2\theta) + i \sin (2\theta)$$

By induction (homework) we have De Moivre's theorem

$$(\cos \theta + i \sin \theta)^n = \cos (n\theta) + i \sin (n\theta)$$

which holds for $n = 0, 1, 2, \dots$

Furthermore (using our earlier result for z^{-1}):

$$\begin{aligned} (\cos \theta + i \sin \theta)^{-1} &= \frac{\cos \theta - i \sin \theta}{\cos^2 \theta + \sin^2 \theta} \\ &= \cos \theta - i \sin \theta \\ &= \cos (-\theta) + i \sin (-\theta) \end{aligned}$$

and so De Moivre's theorem holds for $n = -1, -2, \dots$

De Moivre's theorem has many applications.

Examples

- Trigonometric identities. For example when $n = 2$:

$$\begin{aligned} \cos (2\theta) + i \sin (2\theta) &= (\cos \theta + i \sin \theta)^2 \\ &= \cos^2 \theta + 2i \cos \theta \sin \theta + i^2 \sin^2 \theta \\ &= (\cos^2 \theta - \sin^2 \theta) + i(2 \cos \theta \sin \theta) \end{aligned}$$

Equating the real and imaginary parts separately:

$$\cos (2\theta) = \cos^2 \theta - \sin^2 \theta \quad \text{and} \quad \sin (2\theta) = 2 \cos \theta \sin \theta$$

When $n = 3$:

$$\begin{aligned}\cos(3\theta) + i \sin(3\theta) &= (\cos \theta + i \sin \theta)^3 \\ &= \cos^3 \theta + 3i \cos^2 \theta \sin \theta + 3i^2 \cos \theta \sin^2 \theta + i^3 \sin^3 \theta \\ &= (\cos^3 \theta - 3 \cos \theta \sin^2 \theta) + i(3 \cos^2 \theta \sin \theta - \sin^3 \theta)\end{aligned}$$

where we have used the binomial expansion

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k, \quad \binom{n}{k} = \frac{n!}{(n-k)! k!}$$

with $n = 3$: $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$.

Equating real parts:

$$\begin{aligned}\cos(3\theta) &= \cos^3 \theta - 3 \cos \theta \sin^2 \theta \\ &= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta) \\ &= 4 \cos^3 \theta - 3 \cos \theta\end{aligned}$$

Equating imaginary parts:

$$\begin{aligned}\sin(3\theta) &= 3 \cos^2 \theta \sin \theta - \sin^3 \theta \\ &= 3(1 - \sin^2 \theta) \sin \theta - \sin^3 \theta \\ &= 3 \sin \theta - 4 \sin^3 \theta\end{aligned}$$

Notice how we get two for the price of one.

- Simplifying high powers of a complex number. For example:

$$\begin{aligned}(\sqrt{3} + i)^{18} &= 2^{18} \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i \right)^{18} \\ &= 2^{18} [\cos(\pi/6) + i \sin(\pi/6)]^{18} \\ &= 2^{18} [\cos(18\pi/6) + i \sin(18\pi/6)] \\ &= 2^{18} [\cos(3\pi) + i \sin(3\pi)] = -2^{18}\end{aligned}$$

- Summing series. Consider for example

$$\begin{aligned}\sin \theta + \sin (2\theta) + \cdots + \sin (n\theta) &= \sum_{k=1}^n \sin (k\theta) \\ &= \operatorname{Im} \sum_{k=1}^n \exp (ik\theta)\end{aligned}$$

De Moivre's theorem in "exp" notation reads $\exp (ik\theta) = [\exp (i\theta)]^k$, so above we just have a geometric series:

$$\begin{aligned}\sum_{k=1}^n \exp (ik\theta) &= \sum_{k=1}^n [\exp (i\theta)]^k \\ &= \exp (i\theta) \frac{1 - \exp (in\theta)}{1 - \exp (i\theta)} \\ &= \exp (i\theta) \frac{\exp (in\theta/2)}{\exp (i\theta/2)} \frac{\exp (-in\theta/2) - \exp (in\theta/2)}{\exp (-i\theta/2) - \exp (i\theta/2)} \\ &= \exp [i(n+1)\theta/2] \frac{\sin (n\theta/2)}{\sin (\theta/2)}\end{aligned}$$

Taking real and imaginary parts:

$$\begin{aligned}\cos \theta + \cos (2\theta) + \cdots + \cos (n\theta) &= \operatorname{Re} \sum_{k=1}^n \exp (ik\theta) \\ &= \cos [(n+1)\theta/2] \frac{\sin (n\theta/2)}{\sin (\theta/2)} \\ \sin \theta + \sin (2\theta) + \cdots + \sin (n\theta) &= \operatorname{Im} \sum_{k=1}^n \exp (ik\theta) \\ &= \sin [(n+1)\theta/2] \frac{\sin (n\theta/2)}{\sin (\theta/2)}\end{aligned}$$

11 Integer powers

From $z = |z| \exp(i \arg z)$:

$$\begin{aligned} z^2 &= |z|^2 \exp(i \arg z) \exp(i \arg z) \\ &= |z|^2 \exp(i 2 \arg z) \end{aligned}$$

Therefore $|z^2| = |z|^2$ and $\arg(z^2) = 2 \arg z [+2k\pi]$.

Why is the $[+2k\pi]$ included above?

Similarly:

- $|z^{-1}| = |z|^{-1}$, $\arg(z^{-1}) = -\arg z$
- $|z^*| = |z|$, $\arg(z^*) = -\arg z$
- for n integer $|z^n| = |z|^n$, $\arg(z^n) = n \arg z [+2k\pi]$
- $|z_1 z_2| = |z_1| |z_2|$, $\arg(z_1 z_2) = \arg z_1 + \arg z_2$
- $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$, $\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$
- $|-z| = |z|$, $\arg(-z) = \arg z + \pi$

For example

$$\begin{aligned} z^{-1} = \frac{1}{z} &= \frac{1}{|z| \exp(i \arg z)} \\ &= \frac{1}{|z|} \exp(-i \arg z) \end{aligned}$$

$$\begin{aligned} z_1 z_2 &= |z_1| |z_2| \exp(i \arg z_1) \exp(i \arg z_2) \\ &= |z_1| |z_2| \exp[i(\arg z_1 + \arg z_2)] \end{aligned}$$

12 n -th roots

Complex powers are generally multi-valued (integer powers are the exception). Roots are our first encounter with that. We wish to solve

$$z^n = w$$

for z . First take moduli of both sides:

$$|z|^n = |w| \quad \Rightarrow \quad |z| = \sqrt[n]{|w|}$$

where $\sqrt[n]{|w|}$ denotes the familiar positive real n -th root.

Next write $z = |z| \exp(i\theta)$ and $w = |w| \exp(i \operatorname{Arg} w)$:

$$\begin{aligned} z^n = w &\Rightarrow |z|^n \exp(in\theta) = |w| \exp(i \operatorname{Arg} w) \\ &\Rightarrow \exp(in\theta) = \exp(i \operatorname{Arg} w) \end{aligned}$$

Therefore $n\theta = \operatorname{Arg} w + 2k\pi$, or

$$\theta_k = \frac{\operatorname{Arg} w + 2k\pi}{n}$$

Distinct solutions for z arise by taking n consecutive values of k , e.g.:

$$z_k = \sqrt[n]{|w|} \exp(i\theta_k) \quad , \quad k = 0, 1, \dots, n-1$$

Increasing k further just repeats the solutions for z .

Example

$$z^3 = 1 - i\sqrt{3}$$

Express the RHS in polar form:

$$\begin{aligned} w = 1 - i\sqrt{3} &= |w| \frac{w}{|w|} \\ &= 2 \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) \\ &= 2 [\cos(-\pi/3) + i \sin(-\pi/3)] = 2 \exp(-i\pi/3) \end{aligned}$$

“by inspection”. Or following the “recipe”:

$$\begin{aligned} w &= 1 - i\sqrt{3} \\ \Rightarrow |w| &= \sqrt{1+3} = 2, \quad \text{Arg } w = \text{Arctan}(-\sqrt{3}) = -\pi/3 \end{aligned}$$

There are three distinct solutions for z :

$$z_k = \sqrt[3]{2} \exp(i\theta_k) \quad , \quad \theta_k = \frac{-\pi/3 + 2k\pi}{3} \quad , \quad k = 0, 1, 2$$

$$\begin{aligned} z_0 &= \sqrt[3]{2} \exp(-i\pi/9) = \sqrt[3]{2} \cos(-\pi/9) + i \sqrt[3]{2} \sin(-\pi/9) \\ &= \sqrt[3]{2} \cos(\pi/9) - i \sqrt[3]{2} \sin(\pi/9) \end{aligned}$$

$$z_1 = \sqrt[3]{2} \exp(i5\pi/9) = \sqrt[3]{2} \cos(5\pi/9) + i \sqrt[3]{2} \sin(5\pi/9)$$

$$z_2 = \sqrt[3]{2} \exp(i11\pi/9) = \sqrt[3]{2} \cos(11\pi/9) + i \sqrt[3]{2} \sin(11\pi/9)$$

13 Inequalities

It is not possible to define inequalities between complex numbers. However they exist between the moduli of complex numbers which are real numbers.

Useful examples are the triangle inequalities

$$\begin{aligned} |z_1 + z_2| &\leq |z_1| + |z_2| \\ |z_1 - z_2| &\geq ||z_1| - |z_2|| \end{aligned}$$

To prove this we need

$$|z| \geq |\text{Re } z| \quad \text{and} \quad |z| \geq |\text{Im } z|$$

which both follow from

$$|z|^2 = |\text{Re } z|^2 + |\text{Im } z|^2$$

Consider

$$\begin{aligned} |z_1 + z_2|^2 &= (z_1 + z_2)(z_1 + z_2)^* \\ &= (z_1 + z_2)(z_1^* + z_2^*) \\ &= |z_1|^2 + |z_2|^2 + (z_2 z_1^* + z_1 z_2^*) \\ &= |z_1|^2 + |z_2|^2 + 2 \operatorname{Re}(z_2 z_1^*) \\ &\leq |z_1|^2 + |z_2|^2 + 2|z_2 z_1^*| \\ &= |z_1|^2 + |z_2|^2 + 2|z_2| |z_1| \\ &= (|z_1| + |z_2|)^2 \end{aligned}$$

Take square roots for the triangle inequality.

14 Polynomial equations

These are equations of the form

$$a_0 + a_1z + a_2z^2 + \cdots + a_nz^n = 0 \quad n > 0, \quad a_n \neq 0$$

where $a_0, a_1, \dots, a_n$ are complex coefficients. This equation has at least 1 and at most n distinct solutions for z .

An example is the quadratic equation

$$az^2 + bz + c = 0 \quad , \quad a \neq 0$$

The solutions are

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example

$$z^2 + z + 1 = 0 \quad \Rightarrow \quad z = -\frac{1}{2} \pm i\frac{\sqrt{3}}{2}$$

Example

$$z^3 - 1 = 0$$

We can solve this cubic equation in two ways. First, write it as

$$(z - 1)(z^2 + z + 1) = 0$$

One solution is $z = 1$. The remaining solutions come from setting

$$z^2 + z + 1 = 0$$

and are given above.

The second way is to write the equation as

$$z^3 = 1$$

and solve it by taking the three cubic roots of unity:

$$\begin{aligned} z^3 &= 1 = \exp(2k\pi i) \\ \Rightarrow z_k &= \exp(2k\pi i/3) \quad , \quad k = 0, 1, 2 \end{aligned}$$

which generates the same solutions as above.

15 Complex functions

For each z of a set S the function $f(z)$ defines one or many w in a set T :

$$w = f(z)$$

f is single-valued if there is one w for each z .

f is multi-valued if there are many w for each z .

Both z and w are complex numbers. We can write $z = x + iy$ and $w = u + iv$, where x, y and u, v are real. Then $w = f(z)$ becomes

$$f(z) = f(x + iy) = u(x, y) + iv(x, y)$$

Thus a complex function f can be replaced by a pair of real functions u and v which depend on the two real variables x and y .

Examples

$$\begin{aligned} f(z) = z^2 = (x + iy)^2 &= (x^2 - y^2) + i2xy \\ \Rightarrow u(x, y) = x^2 - y^2 &\quad \text{and} \quad v(x, y) = 2xy \end{aligned}$$

$$\begin{aligned} f(z) = |z|^2 = zz^* &= x^2 + y^2 \\ \Rightarrow u(x, y) = x^2 + y^2 &\quad \text{and} \quad v(x, y) = 0 \end{aligned}$$

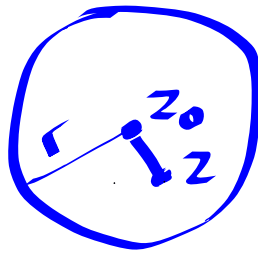
16 Neighbourhood and annulus

Consider complex numbers as points on the complex plane.

- A neighbourhood (open disc) is the set of points z such that

$$|z - z_0| < r$$

These points lie within a circle of radius r and centre z_0 .



- A punctured neighbourhood is the set of points z such that

$$0 < |z - z_0| < r$$

- An annulus is the set of points z such that

$$r_1 < |z - z_0| < r_2$$

These points lie between the concentric circles of radii r_1 and r_2 with centre z_0 .

17 Limit

Let $f(z)$ be a single-valued function defined in a neighbourhood of z_0 . w_0 is the limit of $f(z)$ as z approaches z_0 if for every $\epsilon > 0$ there exists a $\delta > 0$ such that

$$|f(z) - w_0| < \epsilon \quad \text{for all } z \text{ such that} \quad 0 < |z - z_0| < \delta$$

This is written as

$$\lim_{z \rightarrow z_0} f(z) = w_0$$

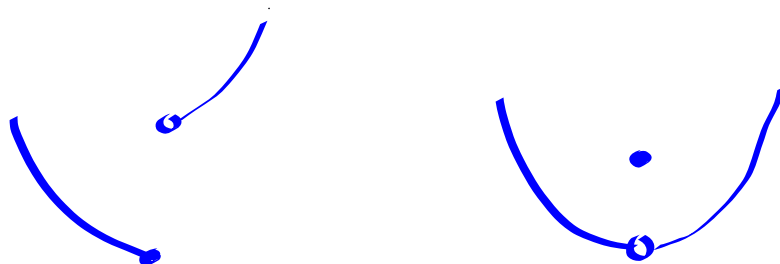
In other words, $f(z)$ can be made arbitrarily close to w_0 by making z sufficiently close to z_0 .

A key feature of the above definition is that the limit, if it exists, should be independent of the direction from which z approaches z_0 . This is the foundation of complex analysis.

18 Continuity

$f(z)$ is continuous at z_0 provided $f(z_0)$ is defined, $\lim_{z \rightarrow z_0} f(z)$ exists, and

$$\lim_{z \rightarrow z_0} f(z) = f(z_0)$$



Examples

- $f(z) = z^2$ is continuous at $z_0 = 0$. To establish this first note that $f(z_0) = f(0) = 0$ is defined. Second, the limit

$$\lim_{z \rightarrow z_0} f(z) = \lim_{z \rightarrow 0} z^2 = 0$$

exists. Third, $\lim_{z \rightarrow z_0} f(z) = f(z_0)$.

- Consider by contrast

$$f(z) = \begin{cases} z^2 & , \quad z \neq 0 \\ 2 & , \quad z = 0 \end{cases}$$

$f(z_0) = f(0) = 2$ is defined. The limit

$$\lim_{z \rightarrow z_0} f(z) = \lim_{z \rightarrow 0} z^2 = 0$$

exists. But $\lim_{z \rightarrow z_0} f(z) \neq f(z_0)$. Therefore $f(z)$ is not continuous at $z_0 = 0$.

If $f_1(z)$ and $f_2(z)$ are continuous then the following are continuous:

- $f_1(z) \pm f_2(z)$
- $f_1(z) f_2(z)$
- $f_1(z)/f_2(z)$ provided $f_2(z) \neq 0$
- $f_1(f_2(z))$

If $f(z)$ is continuous in a bounded closed region R , there exists M such that $|f(z)| < M$ for all z in R .

(A bounded region is one for which $|z| < d$ for all points in the region. A closed region includes the boundary.)

19 Derivative

A function $f(z)$ is differentiable at a point z if the limit

$$\lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

exists.

The derivative of $f(z)$ is defined as this limit:

$$\frac{df(z)}{dz} = f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

So far everything looks similar to real analysis. The key new element is the stringency built into the definition of a limit in the complex plane: that to exist, this limit should be independent of the direction from which Δz approaches 0. This sifts out only very special functions.

The usual rules of differentiation carry over:

$$\begin{aligned}\frac{d}{dz} (f_1 \pm f_2) &= \frac{df_1}{dz} \pm \frac{df_2}{dz} \\ \frac{d}{dz} (f_1 f_2) &= \frac{df_1}{dz} f_2 + f_1 \frac{df_2}{dz} \\ \frac{d}{dz} [f_1(f_2(z))] &= \frac{df_1}{df_2} \frac{df_2}{dz}\end{aligned}$$

Example: differentiation from first principles

We can test for differentiability and evaluate derivatives from scratch.

- Consider $f(z) = az^2$, $a = \text{const} \neq 0$:

$$\begin{aligned}\frac{f(z + \Delta z) - f(z)}{\Delta z} &= a \frac{(z + \Delta z)^2 - z^2}{\Delta z} \\ &= a \frac{2z\Delta z + (\Delta z)^2}{\Delta z} \\ &= a(2z + \Delta z)\end{aligned}$$

Therefore

$$\begin{aligned}\frac{f(z + \Delta z) - f(z)}{\Delta z} - 2az &= a\Delta z \\ \Rightarrow \left| \frac{f(z + \Delta z) - f(z)}{\Delta z} - 2az \right| &= |a||\Delta z|\end{aligned}$$

We can make the left as small as we please by taking Δz sufficiently close to 0, irrespective of the direction of approach. Specifically, for any $\epsilon > 0$ (however small), the left is $< \epsilon$ whenever $0 < |\Delta z| < \epsilon/|a|$.

By the definition of a limit

$$\lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = 2az$$

The function is differentiable, with derivative

$$\frac{d}{dz}(az^2) = 2az$$

Furthermore the above result holds at all z and therefore az^2 is differentiable everywhere.

- Consider by contrast $f(z) = zz^* = x^2 + y^2$:

$$\begin{aligned}\frac{f(z + \Delta z) - f(z)}{\Delta z} &= \frac{(x + \Delta x)^2 + (y + \Delta y)^2 - x^2 - y^2}{\Delta x + i\Delta y} \\ &= \frac{2x\Delta x + 2y\Delta y}{\Delta x + i\Delta y} + \frac{(\Delta x)^2 + (\Delta y)^2}{\Delta x + i\Delta y}\end{aligned}$$

Write Δz in polar form: $\Delta z = |\Delta z| \exp(i\theta)$, where $\theta = \text{Arg}(\Delta z)$. Then $\Delta x = |\Delta z| \cos \theta$, $\Delta y = |\Delta z| \sin \theta$ and

$$\frac{f(z + \Delta z) - f(z)}{\Delta z} = \frac{2x + 2y \tan \theta}{1 + i \tan \theta} + |\Delta z| \exp(-i\theta)$$

The second term vanishes with $|\Delta z|$, but we are left with the first term, which depends on $\theta = \text{Arg}(\Delta z)$ and thus on the direction of approach. Therefore

$$\lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \quad \text{does not exist}$$

and the function is not differentiable, except at $z = 0$. At $z = 0$ the first term is zero, and the limit does exist and equals zero: our function is differentiable at that single point.

20 Analytic functions

If $f(z)$ is differentiable at $z = z_0$ and at every point in a neighbourhood of z_0 then $f(z)$ is analytic (or holomorphic) at z_0 .

If $f(z) = u(x, y) + iv(x, y)$ is analytic then u and v satisfy the Cauchy-Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

The converse is that if the first order partial derivatives of u and v are continuous and satisfy the Cauchy-Riemann equations in an open region then $f = u + iv$ is analytic in the region.

21 Cauchy-Riemann equations

To show that an analytic function $f(z) = u(x, y) + iv(x, y)$ satisfies the Cauchy-Riemann equations consider

$$\frac{df(z)}{dz} = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

In terms of u, v, x, y we have $\Delta z = \Delta x + i\Delta y$ and

$$\begin{aligned} \Delta f &= f(z + \Delta z) - f(z) \\ &= u(x + \Delta x, y + \Delta y) + i v(x + \Delta x, y + \Delta y) - u(x, y) - i v(x, y) \\ &= [u(x + \Delta x, y + \Delta y) - u(x, y)] + i [v(x + \Delta x, y + \Delta y) - v(x, y)] \end{aligned}$$

Choosing a path for which $\Delta y = 0$ gives $\Delta z = \Delta x$ and

$$\Delta f = [u(x + \Delta x, y) - u(x, y)] + i [v(x + \Delta x, y) - v(x, y)]$$

Then

$$\begin{aligned} \frac{df(z)}{dz} &= \lim_{\Delta x \rightarrow 0} \left(\frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x} \right) \\ &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \end{aligned}$$

Choosing a path for which $\Delta x = 0$ gives $\Delta z = i\Delta y$ and

$$\begin{aligned} \frac{df(z)}{dz} &= \lim_{\Delta y \rightarrow 0} \left(\frac{u(x, y + \Delta y) - u(x, y)}{i\Delta y} + i \frac{v(x, y + \Delta y) - v(x, y)}{i\Delta y} \right) \\ &= -i \left(\frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y} \right) \\ &= -i \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \end{aligned}$$

These two expressions for the derivative must be equal. Equating real and imaginary parts gives the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

We will often use the short-hand notation $u_x = \partial u / \partial x$, etc:

$$u_x = v_y \quad \text{and} \quad u_y = -v_x$$

We also get a handy way of evaluating derivatives if they exist. From above, if $f(z)$ is differentiable then

$$\begin{aligned}\frac{df(z)}{dz} &= u_x + \mathrm{i}v_x \\ &= u_x - \mathrm{i}u_y \\ &= v_y - \mathrm{i}u_y \\ &= v_y + \mathrm{i}v_x\end{aligned}$$

Examples

- $f(z) = z^2$ is analytic everywhere. Let $z = x + \mathrm{i}y$ so that $f(z) = (x^2 - y^2) + \mathrm{i}(2xy)$. Therefore $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$. The partial derivatives are $u_x = 2x$, $u_y = -2y$, $v_x = 2y$, $v_y = 2x$. The Cauchy-Riemann equations are satisfied everywhere. The derivative is

$$\frac{d}{dz}(z^2) = u_x + \mathrm{i}v_x = 2x + \mathrm{i}2y = 2z$$

as we have already seen.

- $f(z) = z^*$ is not analytic anywhere. Let $z = x + \mathrm{i}y$ so that $f(z) = x + \mathrm{i}(-y)$. Therefore $u(x, y) = x$ and $v(x, y) = -y$. The partial derivatives are $u_x = 1$, $u_y = 0$, $v_x = 0$, $v_y = -1$. The Cauchy-Riemann equations are nowhere satisfied.
- $f(z) = zz^*$ is not analytic anywhere. Let $z = x + \mathrm{i}y$ so that $f(z) = x^2 + y^2$. Therefore $u(x, y) = x^2 + y^2$ and $v(x, y) = 0$. The partial derivatives are $u_x = 2x$, $u_y = 2y$, $v_x = 0$, $v_y = 0$. The Cauchy-Riemann equations are satisfied only at the point $z = 0$. Analyticity refers to a neighbourhood so the function is nowhere analytic.

Polynomials are analytic everywhere.

Rational functions (the ratio of two polynomials) are analytic except at the zeros of the denominator.

22 Exponential function

We define the exponential function by

$$\exp z = \exp(x + iy) = e^x (\cos y + i \sin y)$$

where e^x is the familiar real exponential function, $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$.

Key properties follow.

- When $y = 0$, $\exp z$ reduces to $\exp x = e^x$. When $x = 0$, we get

$$\exp(iy) = \cos y + i \sin y$$

which is the Euler formula.

- $\exp z$ is analytic:

$$\begin{aligned} \exp z &= e^x \cos y + ie^x \sin y = u + iv \\ u = e^x \cos y &\Rightarrow u_x = e^x \cos y \quad \text{and} \quad u_y = -e^x \sin y \\ v = e^x \sin y &\Rightarrow v_x = e^x \sin y \quad \text{and} \quad v_y = e^x \cos y \end{aligned}$$

and the Cauchy-Riemann conditions are satisfied everywhere.

- The derivative of the exponential function is

$$\begin{aligned} \frac{d}{dz} \exp z &= u_x + iv_x \\ &= e^x \cos y + ie^x \sin y = \exp z \end{aligned}$$

- $|\exp z| = e^x$ and $\arg(\exp z) = y \pmod{2\pi}$.

- We recover the property

$$\begin{aligned} \exp z_1 \exp z_2 &= e^{x_1} e^{x_2} (\cos y_1 + i \sin y_1) (\cos y_2 + i \sin y_2) \\ &= e^{x_1+x_2} [\cos(y_1 + y_2) + i \sin(y_1 + y_2)] = \exp(z_1 + z_2) \end{aligned}$$

- $\exp z$ has no zeros.

An alternative definition of $\exp z$ is by the power series

$$\exp z = \sum_{k=0}^{\infty} \frac{z^k}{k!}$$

which is absolutely convergent for all finite z .

23 Circular functions

The circular functions are defined by

$$\begin{aligned}\cos z &= \frac{\exp(iz) + \exp(-iz)}{2} \\ \sin z &= \frac{\exp(iz) - \exp(-iz)}{2i}\end{aligned}$$

The inverse relations are

$$\exp(\pm iz) = \cos z \pm i \sin z$$

Some properties of these functions are

- $\cos(-z) = \cos z$ and $\sin(-z) = -\sin z$
- $\cos^2 z + \sin^2 z = 1$
- $\cos(z_1 + z_2) = \cos z_1 \cos z_2 - \sin z_1 \sin z_2$
- $\sin(z_1 + z_2) = \sin z_1 \cos z_2 + \cos z_1 \sin z_2$
- $\frac{d}{dz} \cos z = -\sin z$ and $\frac{d}{dz} \sin z = \cos z$
- $\cos(iy) = \cosh y$
- $\sin(iy) = i \sinh y$
- $\cos(x + iy) = \cos x \cosh y - i \sin x \sinh y$
- $\sin(x + iy) = \sin x \cosh y + i \cos x \sinh y$
- $|\cos z| = \sqrt{\cos^2 x + \sinh^2 y}$
- $|\sin z| = \sqrt{\sin^2 x + \sinh^2 y}$
- $\cos z = 0$ when $z = (2k + 1)\pi/2$
- $\sin z = 0$ when $z = k\pi$
- $\cos(z + 2k\pi) = \cos z$ and $\sin(z + 2k\pi) = \sin z$

for integer k .

The power series are

$$\cos z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k}}{(2k)!} \quad \text{and} \quad \sin z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{(2k+1)!}$$

which are absolutely convergent for all finite z .

The remaining circular functions are defined in the usual way, provided the denominators are non-zero:

$$\tan z = \frac{\sin z}{\cos z} \quad \cot z = \frac{1}{\tan z}$$

$$\sec z = \frac{1}{\cos z} \quad \operatorname{cosec} z = \frac{1}{\sin z}$$

24 Hyperbolic functions

The hyperbolic functions are defined by

$$\begin{aligned} \cosh z &= \frac{\exp z + \exp(-z)}{2} \\ \sinh z &= \frac{\exp z - \exp(-z)}{2} \end{aligned}$$

The inverse relations are

$$\exp(\pm z) = \cosh z \pm \sinh z$$

Some properties of these functions are

- $\cosh(-z) = \cosh z$ and $\sinh(-z) = -\sinh z$
- $\cosh^2 z - \sinh^2 z = 1$
- $\cosh(z_1 + z_2) = \cosh z_1 \cosh z_2 + \sinh z_1 \sinh z_2$
- $\sinh(z_1 + z_2) = \sinh z_1 \cosh z_2 + \cosh z_1 \sinh z_2$
- $\frac{d}{dz} \cosh z = \sinh z$ and $\frac{d}{dz} \sinh z = \cosh z$

- $\cosh(iy) = \cos y$
- $\sinh(iy) = i \sin y$
- $\cosh(x + iy) = \cosh x \cos y + i \sinh x \sin y$
- $\sinh(x + iy) = \sinh x \cos y + i \cosh x \sin y$
- $|\cosh z| = \sqrt{\sinh^2 x + \cos^2 y}$
- $|\sinh z| = \sqrt{\sinh^2 x + \sin^2 y}$
- $\cosh z = 0$ when $z = (2k + 1)\pi i/2$
- $\sinh z = 0$ when $z = k\pi i$
- $\cosh(z + 2k\pi i) = \cosh z$ and $\sinh(z + 2k\pi i) = \sinh z$

for integer k .

The power series are

$$\cosh z = \sum_{k=0}^{\infty} \frac{z^{2k}}{(2k)!} \quad \text{and} \quad \sinh z = \sum_{k=0}^{\infty} \frac{z^{2k+1}}{(2k+1)!}$$

which are absolutely convergent for all finite z .

The remaining hyperbolic functions are defined in the usual way, provided the denominators are non-zero:

$$\begin{aligned} \tanh z &= \frac{\sinh z}{\cosh z} & \coth z &= \frac{1}{\tanh z} \\ \operatorname{sech} z &= \frac{1}{\cosh z} & \operatorname{cosech} z &= \frac{1}{\sinh z} \end{aligned}$$

25 Entire functions

Functions that are analytic at all finite points of the complex plane are called entire.

Now familiar examples of entire functions are: polynomial, exponential, sine and cosine, hyperbolic sine and cosine.

Liouville's boundedness theorem states that a bounded entire function must be constant.

26 Logarithm

The logarithm is the inverse of the exponential function:

$$\exp w = z \quad \Leftrightarrow \quad w = \log z$$

- $\exp w \neq 0$ so $\log 0$ is not defined.
- We use $\log$ for the natural logarithm (sometimes written as $\ln$).

We define

$$\log z = \log |z| + i \arg z \quad (z \neq 0)$$

where the first term on the right is the familiar real logarithm.

- To check, take the exponential of the RHS:

$$\exp(\log z) = \exp(\log |z|) \exp(i \arg z) = |z| \exp(i \arg z) = z$$

- $\log z$ is many-valued:

$$\begin{aligned} \log z &= \log |z| + i \arg z \\ &= \log |z| + i (\text{Arg } z + 2k\pi) \quad \text{where} \quad k = 0, \pm 1, \pm 2, \dots \end{aligned}$$

- The principal value (denoted $\text{Log } z$) corresponds to choosing $\text{Arg } z$ for the argument:

$$\text{Log } z = \log |z| + i \text{Arg } z \quad (-\pi < \text{Arg } z \leq \pi)$$

Examples

$$\begin{aligned} \log(-1) &= i(\pi + 2k\pi) \quad , \quad k = 0, \pm 1, \pm 2, \dots \\ \text{Log}(-1) &= i\pi \quad , \quad \text{Log}(-2i) = \log 2 - i\pi/2 \end{aligned}$$

- $\log(z_1 z_2) = \log z_1 + \log z_2$ and $\log(z_1/z_2) = \log z_1 - \log z_2$

As each side is multi-valued, the equations mean that every entry in one list appears also in the other.

- $\frac{dz}{dw} = \exp w = z \quad \Rightarrow \quad \frac{dw}{dz} = \frac{d}{dz} \log z = \frac{1}{z}$

When $|z| \leq 1$ and $z \neq -1$

$$\log(1+z) = z - \frac{1}{2}z^2 + \frac{1}{3}z^3 - \dots$$

For example $\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$

27 Complex powers

We define complex powers as:

$$z^w = \exp(w \log z)$$

In general, this is a multi-valued function of z .

Examples

- $i^i = \exp(i \log i)$

$$\begin{aligned} \log i &= \log|i| + i \arg i \\ &= i(\pi/2 + 2k\pi) \quad , \quad k = 0, \pm 1, \pm 2, \dots \\ i^i &= \exp[-(\pi/2 + 2k\pi)] \quad , \quad k = 0, \pm 1, \pm 2, \dots \end{aligned}$$

- $1^i = \exp(i \log 1)$

$$\begin{aligned} \log 1 &= \log|1| + i \arg 1 \\ &= i2k\pi \quad , \quad k = 0, \pm 1, \pm 2, \dots \\ 1^i &= \exp(-2k\pi) \quad , \quad k = 0, \pm 1, \pm 2, \dots \end{aligned}$$

- $(-1)^{(1/\pi)} = \exp[(1/\pi) \log(-1)]$

$$\begin{aligned} \log(-1) &= \log|-1| + i \arg(-1) \\ &= i(\pi + 2k\pi) \quad , \quad k = 0, \pm 1, \pm 2, \dots \\ (-1)^{(1/\pi)} &= \exp(2k+1)i \\ &= \cos(2k+1) + i \sin(2k+1) \quad , \quad k = 0, \pm 1, \pm 2, \dots \end{aligned}$$

28 Branch points and cuts

Branch cuts are a way of modifying the complex plane, so as to make a multi-valued function single-valued.

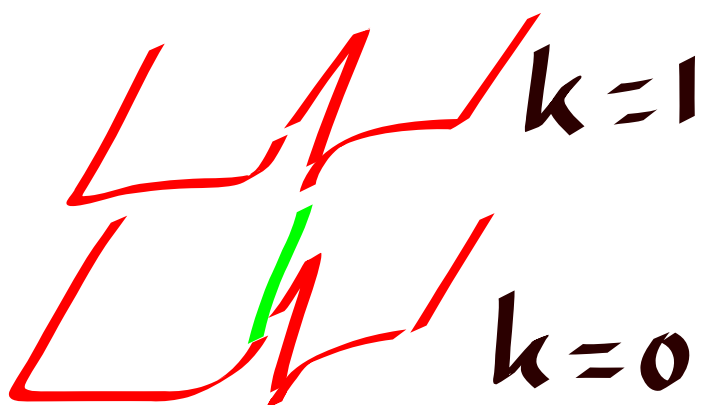
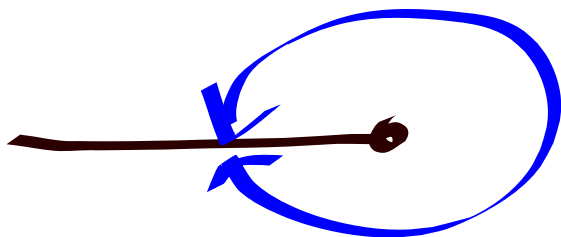
Consider the multi-valued function $f(z) = \log z$. Allow z to make a complete circuit (counter-clockwise) around the origin starting at a point P . At the start we have $z = r \exp(i\theta)$ and $f = \log r + i\theta$. At the end of the circuit $f = \log r + i(\theta + 2\pi)$. After k circuits $f = \log r + i(\theta + 2k\pi)$. We return to the same point P every time but f does not.

We say that the function has different branches labelled by k . For $-\pi < \theta \leq \pi$ we are on the branch $k = 0$; for $\pi < \theta \leq 3\pi$ we are on the branch $k = 1$, etc. Each branch is single-valued.

To maintain the one-valued property we introduce an artificial barrier along the negative real axis including $z = 0$. This is called a branch cut. $z = 0$ is a branch point.

Note that a circuit around any other point does not lead to different values. Thus $z = 0$ is the only branch point.

This is only the first step in a more powerful construction - the Riemann surface.



29 Harmonic functions

One of the great applications of analytic functions is in generating solutions to Laplace's equation. We can see this from the Cauchy-Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad (\text{a}) \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad (\text{b})$$

Differentiating (a) w.r.t. x and (b) w.r.t. y :

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y} \qquad \frac{\partial^2 u}{\partial y^2} = -\frac{\partial^2 v}{\partial y \partial x}$$

Adding the two:
$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Differentiating (a) w.r.t. y and (b) w.r.t. x :

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 v}{\partial y^2} \qquad \frac{\partial^2 u}{\partial x \partial y} = -\frac{\partial^2 v}{\partial x^2}$$

Subtracting the two:
$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

Thus any analytic function automatically provides two solutions to Laplace's equation. Furthermore:

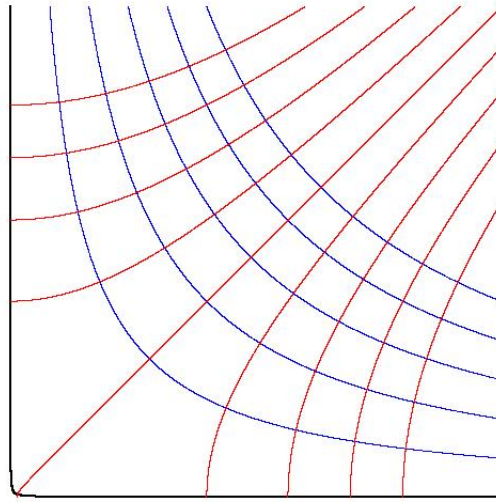
$$\begin{aligned} \nabla u \cdot \nabla v &= u_x v_x + u_y v_y \\ &= u_x(-u_y) + u_y(u_x) = 0 \end{aligned}$$

Thus the families of curves $u = \text{const}$ and $v = \text{const}$ are orthogonal. Such pairs of harmonic functions are called harmonic conjugates.

Laplace's equation plays a fundamental role in electrostatics and fluid flow. For example consider

$$f(z) = z^2 = u + iv = (x^2 - y^2) + i2xy$$

The curves $v = 2xy = \text{const}$ give the equipotentials between two charged plates meeting at right angles, while $u = x^2 - y^2 = \text{const}$ are the electric field lines. The curves $v = 2xy = \text{const}$ give the streamlines for irrotational incompressible fluid flow past a 90-degree bend.



30 Series

Functions such as $\exp z$ have an infinite power series. In fact power series provide an alternative definition of functions.

For series with an infinite number of terms we must define convergence.

Consider a general series

$$f_1(z) + f_2(z) + \cdots + f_n(z) + \cdots$$

where each term is a function of z .

Define the partial sums

$$S_n(z) = f_1(z) + f_2(z) + \cdots + f_n(z)$$

The series converges to a value $S(z)$, provided the sequence of partial sums converges to $S(z)$, i.e. provided $\lim_{n \rightarrow \infty} S_n(z)$ exists and equals $S(z)$:

$$\lim_{n \rightarrow \infty} S_n(z) = S(z)$$

Formally: given z and any $\epsilon > 0$, we can find N such that

$$|S_n(z) - S(z)| < \epsilon \quad \text{for all } n > N$$

- The values of z for which the series converges define the region of convergence.
- A series which is not convergent is divergent.
- The series $f_1(z) + f_2(z) + \cdots$ converges absolutely if $|f_1(z)| + |f_2(z)| + \cdots$ converges.
- Absolute convergence implies convergence.

An absolutely convergent series can be rearranged without affecting the value of the series. Absolutely convergent series can be multiplied.

31 Convergence tests

There are various tests for convergence. We consider the two most familiar ones.

• Comparison test

Suppose the series $g_1(z) + g_2(z) + \dots$ converges absolutely, and $|f_n(z)| \leq |g_n(z)|$ (for all n beyond some N). Then the series $f_1(z) + f_2(z) + \dots$ is absolutely convergent.

• Ratio test (d'Alembert)

Let

$$\xi = \lim_{n \rightarrow \infty} \left| \frac{f_{n+1}(z)}{f_n(z)} \right|$$

If $\xi < 1$ the series $f_1(z) + f_2(z) + \dots$ is absolutely convergent;

if $\xi > 1$ the series diverges;

if $\xi = 1$ we must investigate further.

Examples

- Consider the geometric series

$$S(z) = 1 + z + z^2 + \dots$$

Form the ratio

$$\left| \frac{f_{n+1}(z)}{f_n(z)} \right| = \left| \frac{z^n}{z^{n-1}} \right| = |z|$$

and we have absolute convergence for $|z| < 1$, and divergence for $|z| > 1$.

Now consider the boundary $|z| = \xi = 1$:

for $z = 1$ the partial sums grow without limit;

for $z = \exp(i\theta)$ with $0 < \theta < 2\pi$ we have oscillatory behaviour, which is a form of non-convergence.

The partial sums are

$$S_n(z) = \frac{1 - z^n}{1 - z}$$

and for $|z| < 1$

$$S(z) = \lim_{n \rightarrow \infty} S_n(z) = \frac{1}{1 - z}$$

• Consider

$$S(z) = \sum_{n=1}^{\infty} \frac{n(n+1)}{3^n} z^n$$

Form the ratio

$$\left| \frac{f_{n+1}(z)}{f_n(z)} \right| = \left| \frac{(n+2)z}{3n} \right|$$

and

$$\xi = \lim_{n \rightarrow \infty} \left| \frac{(n+2)z}{3n} \right| = \frac{|z|}{3}$$

Hence we have absolute convergence for $|z| < 3$.

• Consider

$$S(z) = \sum_{n=1}^{\infty} \frac{1}{n^2} \left(\frac{z+1}{z-1} \right)^{n-1}$$

Form the ratio

$$\left| \frac{f_{n+1}(z)}{f_n(z)} \right| = \left| \frac{n^2(z+1)}{(n+1)^2(z-1)} \right|$$

and

$$\xi = \lim_{n \rightarrow \infty} \left| \frac{n^2(z+1)}{(n+1)^2(z-1)} \right| = \left| \frac{z+1}{z-1} \right|$$

Hence we have absolute convergence for

$$\left| \frac{z+1}{z-1} \right| < 1 \quad \Rightarrow \quad |z+1| < |z-1|$$

Since $|z - 1|$ is the distance from z to 1 and $|z + 1|$ is the distance from z to -1 , this condition corresponds to the set of points which are closer to -1 than to 1. This is the left half-plane $\operatorname{Re} z < 0$.

The boundary $\left| \frac{z + 1}{z - 1} \right| = 1$ is the imaginary axis ($\operatorname{Re} z = 0$). Then the series of absolute terms is

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots$$

which is convergent. The region of absolute convergence includes the boundary in this case.